23JMO1

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Problem

Find all triples of positive integers (x, y, z) that satisfy the equation

$$2(x+y+z+2xyz)^2 = (2xy+2yz+2zx+1)^2 + 2023.$$

Solution. Rearranging, we have, $-2023 = (2xy+2yz+2zx+1)^2 - 2(x+y+z+2xyz)^2$. This is a Pell's equation of the form $a^2 - 2b^2$.
We know that this factorises into $(a-\sqrt{2}b)(a+\sqrt{2}b)$. This motivates us to factorise the given expression in a similar fashion.
Observe that, $(2xy+2yz+2zx+1)+\sqrt{2}(x+y+z+2xyz) = (1+\sqrt{2}x)(1+\sqrt{2}y)(1+\sqrt{2}z)$.
Similarly, $(2xy+2yz+2zx+1)-\sqrt{2}(x+y+z+2xyz) = (1-\sqrt{2}x)(1-\sqrt{2}y)(1-\sqrt{2}z)$.
Combining this together, we have, $(1-2x^2)(1-2y^2)(1-2z^2) = -2023$. This implies, $(2x^2-1)(2y^2-1)(2z^2-1) = 2023 = 7 \cdot 17^2$.
Since $x, y, z \in \mathbb{Z}$, checking gives us that $(x, y, z) = (2, 3, 3), (3, 2, 3), (3, 3, 2)$ are the only working triplets.